

MATRIX HEADLIGHT

Intelligent Matrix Headlight for Night Vision

1. Project Overview

Intelligent Matrix Headlight for Night Vision is an embedded-system project designed to improve nighttime driving visibility while reducing glare for oncoming vehicles.

The system uses LDR sensors, Arduino, LED matrix lights, and PWM control to automatically adjust the intensity of individual lights based on the surrounding light conditions.

Goal: To create a smarter and safer vehicle lighting system that adapts to nighttime driving conditions.

2. How It Works

Light Detection → Arduino Processing → PWM Control → Adaptive LED Output

- LDR sensors detect the surrounding light conditions.
- Arduino processes the sensor readings.
- PWM is used to control LED brightness.
- Individual sections of the matrix headlight can be adjusted according to the detected conditions.
- This helps maintain road visibility while reducing unnecessary glare.

3. Technologies Used

Arduino · LDR Sensors · LED Matrix · PWM · Embedded C/C++ · Sensor-Based Control

4. Key Result

The prototype demonstrated approximately 80% reduction in glare under the tested conditions.

The project provided practical experience in embedded programming, sensor integration, PWM control, and hardware prototyping.

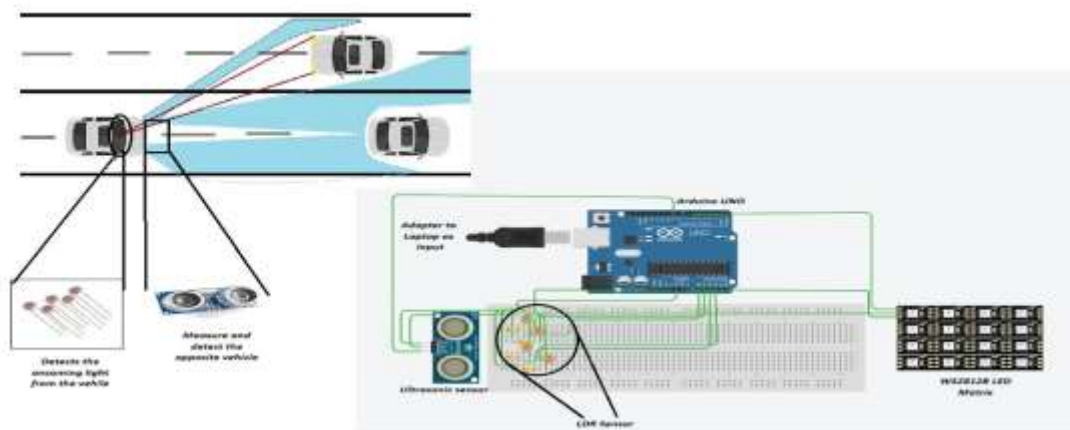
5. Applications

- Adaptive automotive headlights
- Night driving safety
- Automatic brightness control
- Smart vehicle lighting
- Automotive embedded systems

6. Future Scope

The system can be further enhanced using camera-based vehicle detection, automatic beam steering, advanced LED matrices, and AI-based object detection to create a more sophisticated adaptive headlight system.

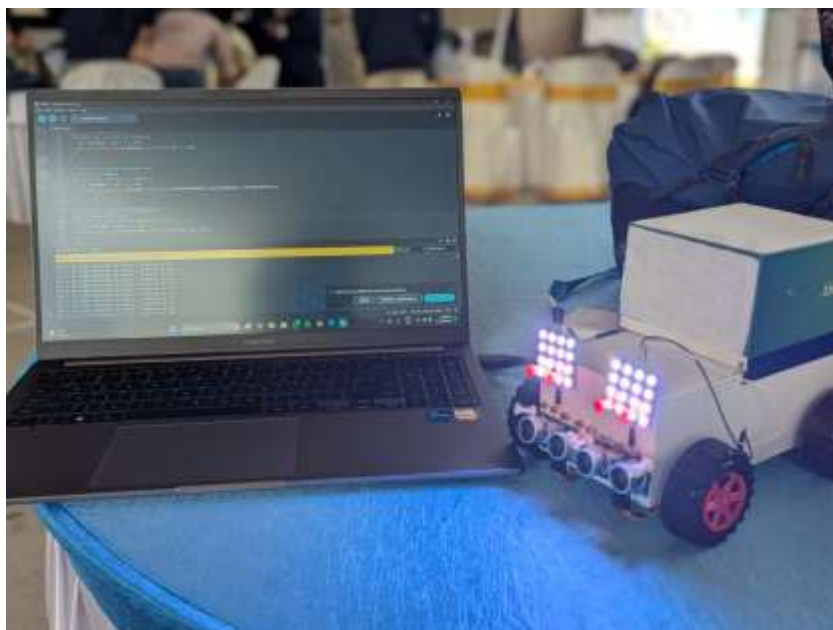
APPENDIX



Methodology



Model Implementation



Model